

FitLens: Efficient Body Measurement Using Camera Vision for Personalized Apparel, Fitness, and E-Commerce

Abstract

One of the biggest problems on e-commerce platforms, or in online apparel, is the high return rates caused by either poor fit or users entering incorrect size information [1], [15]. Users often guess the wrong clothing size due to a lack of knowledge about different brands' sizing [19]. Most online size charts are based on traditional methods of measuring the body with a tape measure [1]. The traditional methods are error-prone and require the physical presence of the user [3], [4]. This is simply not scalable because there are millions of online customers. Some of the other digital solutions, such as 3D scanners or depth cameras, are not scalable since they either require expensive hardware or are not precise enough for clothing size decisions [11], [14]. To solve the major problems with current body measurement systems, we built FitLens AI, a camera-vision-based body measurement system that can extract body measurements from standard 2D photos of both the front and side views without the need for any specialized hardware or accessories.

To get started, the user simply uploads two standard photos: a front view and a side view. Using deep learning algorithms - YOLOv8n-seg for human bounding box detection and segmentation, MediaPipe Pose for extraction of anatomical landmarks, MediaPipe Face Mesh for extraction of precise facial Landmarks. FitLens AI automatically identifies landmark points using MediaPipe from parts of the body for measurements, or gives the user control to manually mark the landmark points on the image and select which point-to-point measurement is needed [13]. Fitlens AI uses YOLOv8 for human detection and MediaPipe Pose for 2D landmark extraction [13], [21]. Data validation and thresholds are essential for the preservation of accuracy during the process. For any input that is deemed to have low detection probability, it will be automatically discarded as part of the immediate error-handling process. The product also uses a multi-stage computer vision pipeline built on a React frontend that interfaces with a Flask backend. This system uses pixel-to-scale conversion.

Pixel distances are first converted into centimeters in the code and then mapped to the sizes of clothing that a user can select from, such as shirt sizes. These measurements include shoulder width, arm length, torso length, hip width, and leg length. In addition to this, asynchronous photo analysis, FitLens AI features a live camera approach, which can capture photos directly from a real-time camera input. The system works in real-time by video capturing 30 frames per second (FPS). The process of analyzing each frame ensures constant monitoring of the body's position.

The workflow utilizes the same YOLOv8 and Mediapipe Pose models for real-time pose estimation and landmark detection [13], [21]. As the user adjusts their posture, the landmarks on their body move with them instantly.

In order to make the best use of this scenario, feedback can be obtained through an instant posture correction mechanism as well as multi-frame averaging of measurements to minimize any pixel instability. In addition to this, face verification is performed to ensure that the same

person is always involved in the process. The product is unique because it doesn't require depth sensors or hardware [8]. FitLens AI offers an instant, scalable, and privacy-driven solution for virtual fitting rooms, personal apparel size determination, and body fitness tracking [8].

Introduction

The global e-commerce apparel market continuously suffers from very high return rates, often exceeding 30%, which is highly driven by improper fitting of clothes, making it an unsustainable trend. This represents a multi-billion-dollar problem, which is a logistical burden and also has a huge environmental impact, as returned items contribute to increased carbon emissions from shipping waste, packaging, and also frequent disposal of unsellable merchandise [1],[15].

In addition to that, the fast growing industry of the digital fitness industry, along with the rising use of telemedicine, has boosted the demand for efficient ways of measuring the human body remotely and accurately. To analyse and evaluate physiological progression, fitness enthusiasts require tools that would measure their body size constantly [2].

Historically, these sectors relied on manual anthropometry, mainly using measuring tapes, which is a time-consuming and inherently inconsistent process, and is labor-intensive and needs the physical presence of trained operators. Studies show that manual measurements may vary depending on the skills of the person who is measuring, the tape tension, posture of the subject, and inter-observer variability, which often exceeds acceptable tolerances for garment fit assessment [4].

Solving body measurement challenges has historically required manual intervention, which may or may not be accessible to average users, and therefore, there is a need for technology-based approaches. While industrial-grade 3D scanners can deliver high accuracy within 1-2cm, they are still very expensive, with the cost ranging from \$10,000 up to \$150,000. Such hardware is unaffordable to average consumers and most retail organizations [7],[14]. This created a gap between the need for accurate body measurements and the availability of consumer-accessible solutions.

With the recent advancements in Computer vision and artificial intelligence, particularly in domains of 2D pose estimation and human key point detection, it seems that a new standard has emerged where spatial data is being extracted directly from digital images [9],[10]. Most of the existing and modern pose estimation methods rely on the use of Convolutional Neural Networks (CNNs) for detecting landmarks in 2D space using a camera without requiring any depth sensors [13]. Frameworks like OpenPose have already shown that real-time key point detection in 2D space can be achieved on standard consumer hardware, making it easy to extract a person's body geometry from the user-provided image data [9]. Similarly, MediaPipe is an example of an ultra-lightweight and cross-platform approach for keypoint detection from standard RGB videos, which is optimized for edge devices and mobile applications [13]. Different versions of the YOLO object detection architecture, including the latest version YOLOv8, offer reliable person detection and extract bounding boxes as part of image pre-processing [21].

Even with the availability of all these tools, building a system that is lightweight, accurate, and works on a standard camera remains an unsolved problem. However, approaches that solely use skeletal joint detection exhibit fast and easy computational efficiency, and still, their ability to measure transverse widths, such as chest width, waist circumference, and hip dimensions, is limited. This is mainly because of loose clothing, which obscures the body boundaries, and the inability of depth estimation using a 2D camera [6][15]. Such types of systems can efficiently detect the internal coordinates of joints, but frequently mispredict the volume of the body, especially when subjects wear non-tight-fitting clothes[12]. Skeletal joint detection-based systems that use only joints as markers work very efficiently with respect to processing time, while being quite computationally inexpensive. However, their accuracy suffers from obstructions caused by loosely fitting clothes as well as the depth ambiguity of monocular images[6],[15]. Although these systems successfully estimate and detect internal joint locations, they do not accurately recognize external volumetric limbs, which makes significant errors when users wear non-fit clothes [12].

Simultaneously, deep learning models, which perform dense estimation and instance segmentation, provide perfect contour mapping and volumetric analysis. The main issue with that is that they require considerable GPU power, thereby prohibiting real-time operations on average consumer hardware and raising a serious issue of energy and heat dissipation on mobile platforms [8],[17]. Currently, there is an ongoing problem of calibrating 2D pixel measurements to real-world measurements, the main reason being lens distortions, perspective changes, pose variety, and lack of calibration protocols [3], [5].

The motivation behind FitLens AI is to address this challenge with a proposed architecture that will enable accurate measurements of body circumferences and dimensions using only 2D camera vision, keypoint landmarks detection, and dynamic pixel-to-scale calibration. For this purpose, the system not only relies on image processing but also provides an extremely effective camera setting that enables real-time video streaming at 30 FPS. Every single frame is examined for the purpose of ensuring continuous pose detection during the process of data collection. The use of multi-frame averaging becomes essential due to the unpredictable nature of the video for stabilizing landmark detection. Through eliminating the need for extensive computations associated with building a complete 3D model of the environment, FitLens AI aims to provide a solution that is fast, portable on consumer-grade mobile devices, and protects users' privacy. The algorithm incorporates light-weighted instance segmentation techniques to handle clothing occlusions and uses heuristic plane-based 2D-to-3D parametric models to solve depth ambiguity, achieving sub-centimeter accuracy while working in offline mode only. Apart from this, for the purpose of ensuring accuracy in the measurement process throughout the continuous monitoring process, a lightweight face recognition element accompanies the pose estimation element. This would ensure that any anomaly in the frame sequence resulting from any changes or interventions is instantly recognized and disregarded. FitLens AI presents a novel approach to automated human body measurement by addressing the inconsistencies in skeletal-only approaches and complex deep neural network frameworks by separating the skeletal joint detection process from morphological contours estimation.

Designed for efficient computational costs, the system is completely edge-driven to ensure that all users' biometrics remain secure in accordance with strict data privacy laws like GDPR and CCPA, while simultaneously providing real-time feedback for correct measurement postures [8].

The core goals of the FitLens AI project can be stated as follows. Firstly, creating and deploying a multi-stage computer vision pipeline capable of performing person detection and bounding-box localization with YOLOv8, extracting 2D anatomical keypoints with MediaPipe Pose, and computing accurate measurement values based on geometric analysis performed in the backend with Python Flask, using React frontend technology. Secondly, introducing the concept of the new pose alignment verification algorithm that would provide real-time visual feedback, allowing users to adopt a standardized measurement pose, thus minimizing pose variability as a source of measurement errors and increasing inter-user consistency. In this feedback loop, what makes it more effective is the live video feed coming from the camera, which enables instant visual feedback like “Stand up straight,” immediately following the movement of the person, along with proper filtering processes that reject any input with less than adequate confidence levels. Thirdly, computing accurately the following anthropometric measures: shoulder width, arm length, torso length, hip width, chest width, and leg length by means of 2D landmark coordinates extraction, dynamic pixel-to-scale mapping, and geometric analysis, taking camera distortions and changes in perspective into account. In order to validate the algorithm, self-consistency tests based on anthropometric data for the body parameters are conducted; any values that do not meet these criteria are automatically discarded before further processing. Fourthly, obtaining an exact geometric error margin not exceeding 1–3 cm when compared to the ground truth values measured manually in all tested dimensions, while significantly outperforming current vision-based approaches with their margin errors in the range of ± 4 –8 cm. Fifthly, implementing the project in accordance with privacy-first principles without any data storage and only with the use of offline. One of the key features of FitLens AI is being able to create the means for the extraction of anthropometric measurements, which ensures consistency and computational efficiency, accurate measurements, and user privacy. Therefore, the technology used provides an efficient and transparent way of doing what proprietary systems cannot accomplish.

Existing Body Measurement Techniques

Body measurements have been taken in many ways over the years, from using a simple measuring tape and size chart to more advanced systems like 3D scanning systems that are too expensive for most people. With so many options available, each having its own advantages and disadvantages, there is a clear need for a better solution. One that is easy to use, accurate, and affordable for everyday users. This is the gap that FitLens aims to address. To understand the importance of FitLens, first, it is useful to look at the existing measurement methods, how they work, and the limitations they have.

1. Traditional and Manual Measurement Methods

When it comes to measuring the human body, the most common method is still the simple measuring tape. In the tailoring industry, this measuring tape tool has been used for many years. ASTM and ISO are the standards followed by these traditional methods, and are performed by trained people who know how to find the correct body points while making sure the person stands during measurement [1].

However, getting proper results every time is not always easy. Little things can affect the measurements, such as stretched tape, lack of concentration, or getting tired from taking measurements of many people. These kinds of small issues can lead to noticeable errors over time.

In most retail stores, measurements are often taken by staff who may not have proper training. This leads to results that do not match the customer's actual body size. Another reason that makes this process less convenient is that some people feel uncomfortable when someone else takes their measurements.

Human error is another common issue. The final result depends on how it is being used, even though the tool is correct. For example, holding the tape too tight or too loose, or reading the value incorrectly, can lead to wrong measurements [3].

It is also difficult to keep the same tape tension every time. As a person continues measuring, tiredness can affect their focus, and identifying the correct body points becomes harder. These small mistakes can slowly reduce the overall accuracy.

This problem becomes even more noticeable in environments like the retail industry. Measurements taken by different people can vary, sometimes by around five centimetres [10]. Changes in posture and measuring style also make it harder to maintain consistency.

In manual measurement, the physical presence of a person is mandatory. While shopping online, people often try to measure themselves at home, which results in less accurate measurements, as it is difficult for one to take their own measurements. It can be difficult to position the body correctly, and without proper guidance, results may not be accurate.

Overall, these issues show that traditional manual measurement methods are not easy to scale. They highly depend on the person taking the measurement and are affected by many small factors that can lead to errors [1], [7].

2. Computer Vision and Depth-Based Measurement Systems

After noticing the limitations of manual methods, researchers started exploring more advanced technologies to find better solutions. Depth cameras and computer vision seemed like a good option at first, but they also had several practical problems [11]. Regular users or

small businesses find it a little difficult to afford these systems, as they are expensive. This consumes our time as it might take at least ten to sixty seconds to complete a scan; they also need a controlled setup to work properly and require an experienced person to operate it [12]. This process may seem a little complex to people who need only the right clothing size.

Because of this, researchers began focusing on simpler and more accessible methods using regular cameras. Pose detection technologies like MediaPipe and OpenPose are very useful and have shown good results in this area [9], [10], [13]. Here, these technologies use deep learning to detect body points without needing any physical markers. Using a regular smartphone camera, MediaPipe can detect up to thirty-three body keypoints in real time [13]. This makes the process much easier and more accessible for everyday users.

As artificial intelligence got better, more of these solutions appeared. A wide range of systems now exist that analyze standard RGB images or video footage to estimate body dimensions, identifying landmarks like the shoulders, hips, and waist to make predictions about a person's measurements. Most of these rely on deep learning frameworks and are designed to work with high-definition images[9].

Among the more widely recognized approaches are CNN- based models, which made meaningful contributions to the field of body pose estimation[10]. MediaPipe is a popular tool that uses AI on phones to find 33 body keypoints from a regular video[13]. OpenPose helps in joint detection from still images[9]. What makes some of these systems especially worthy is their ability to perform all of this in real time, using nothing more than a single video camera feed.

Differences Between Approaches and Their Limitations

The methods used to capture and measure fashion garments for e-commerce have come a long way in recent years. Most of the available systems still have practical issues; even with all the improvements, they limit their use in everyday situations. The need for special hardware is one of the most common problems. Many advanced systems depend on expensive devices like depth cameras and 3D scanners, which makes them hard for regular users or small businesses to afford [7], [14]. Even image-based methods may seem easier to use, but they also have their own challenges. Small factors like camera angle, lighting, and converting image data into actual measurements can reduce accuracy [3], [15].

Apart from hardware, these systems often need controlled lighting, multiple cameras, and trained people to operate them properly [12]. This setup might be difficult for small retailers to manage. Small changes in posture, incorrect positioning, or any body parts not clearly visible in the image can reduce accuracy [3]. And many of the systems do not even check properly if the user is standing correctly while the image is being taken.

Privacy is another important concern. Body measurements can be sensitive personal information; many systems share this information over the internet, which makes the user

hesitant to use the system. This raises concerns about data safety and rules like GDPR, CCPA, and HIPAA, which control how such data should be handled [8].

From a technical point of view, many 2D systems work using pixel values instead of real-world measurements. Using a calibration step, these values will be converted into units like centimetres [5]. Final results get affected if there is an error during this conversion. In addition, many of these systems come with yearly costs, usually between \$500 and \$5,000, which makes it difficult for smaller businesses to afford them [14].

Overall, these problems show that most existing systems require a compromise between accuracy, cost, and ease of use. It is difficult to achieve all three at the same time, which highlights the need for a simpler and more practical solution.

Proposed System Architecture of FitLens AI

System Overview

FitLens AI is designed as an end-to-end computer vision system that automates the measurements of the human body from standard 2D photographs, RGB images, and video streams. RGB images are standard digital photographs captured every day with cameras, such as smartphones or webcams. The AI is built entirely on 2D analysis, so the system works on budget phones, older models, and basic webcams, as it only needs a standard camera. Explicitly avoids expensive 3D reconstruction and depth estimation, since they require a lot of computational power and drain the battery quickly[2], [8]. The architecture follows a three-tier web in which computation is separated across a user-facing React frontend, a Python Flask backend API, and a modular computer vision pipeline[21]. The user interacts with the frontend, uploads two photos - front view and side view, or streams a live camera feed, and receives a structured output of body measurements of 2D keypoints visualization. Processing live video streams introduces a sequence within the video processing workflow. It automatically captures once the user fits within the frame. Running at 30 FPS for approximately 3 seconds, the live camera mode captures frames in real time and processes them frame by frame to ensure temporal consistency and pose tracking throughout the video sequence. The flask backend receives image data through a REST API and coordinates the sequence of processes. Human image is detected by YOLOv8, then the anatomical landmark points are identified by the MediaPipe [13], [21]. The physical measurements are calculated using pixel-to-scale conversion math. Once the math is done, the results are structured as JSON. The central automated workflow of FitLens is that it ingests regular RGB images, MediaPipe processes them to identify human anatomical keypoints, such as the center of the shoulder, elbow, or ankle, and compares the object against a reference calibration, such as the user's known height [13]. The system calculates the ratio between the pixels on the screen and the real-world measurements, such as centimeters. Once the pixel-to-scale conversion is complete, the system calculates the final body dimensions and provides the user with their actual physical measurements. Before the final measurement, the system has a lightweight 2D

face verification algorithm to validate the identity of the user across multiple frames. The face detection protocol ensures that the same person is being measured. The system incorporates five distinct technical workflows regarding FitLens: (1) Capture of real-time images from built-in cameras.

(2) Data refinement - includes frame normalization and resizing.

(3) Pose Estimation through Deep Learning - Uses models like MediaPipe for 2D landmark identification [13].

(4) Quality assurance protocol - If the user is standing too close to the camera or turned sideways, the measurements will be inaccurate

(5) Geometric scaling - Final conversion from pixel-to-scale values[1], [3]. The pipeline is built as a series of discrete Python modules; each part of the system has a strict deal with others.

Facilitating modular validation, easy to replace, and future extension is possible

Flowchart: FitLens AI System Pipeline

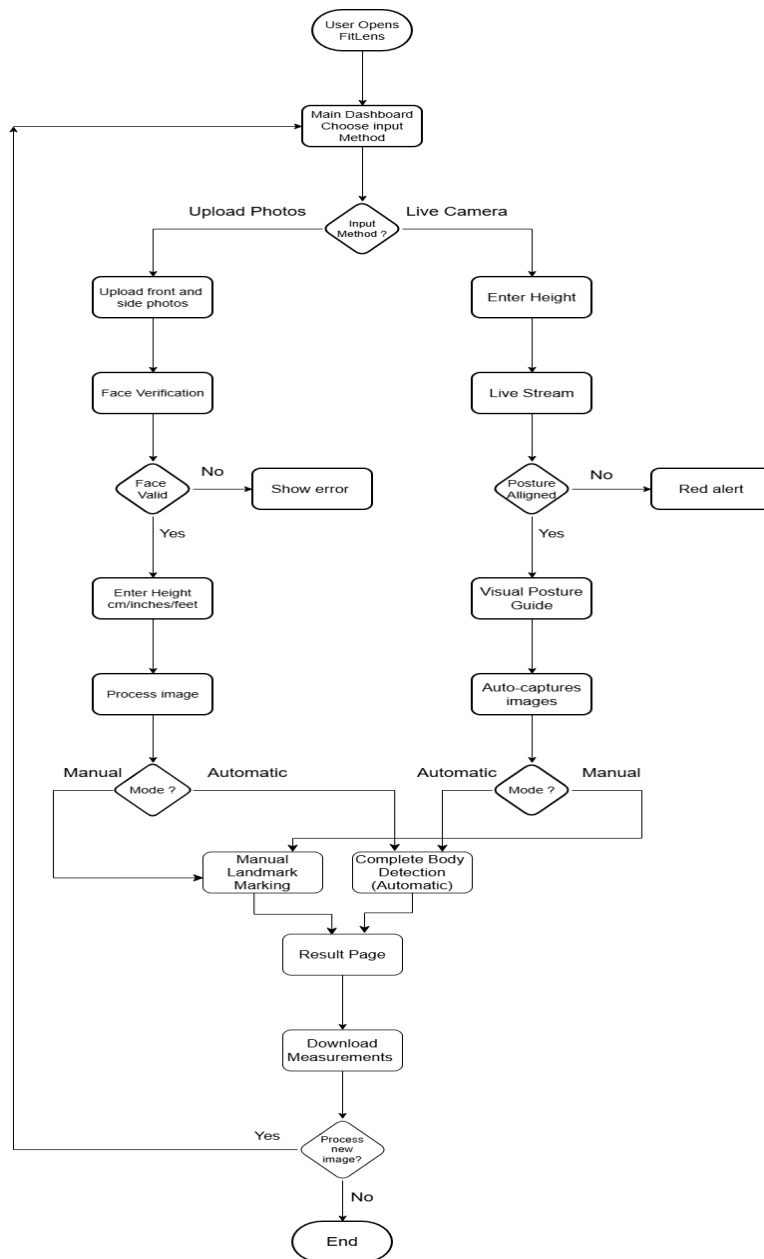


Figure 1: The above Flow Chart shows the entire journey of using FitLens, including upload-photo and live-camera inputs, face verification, posture feedback, manual and automatic measurements, and final output.

A novel feedback mechanism provides real-time correction, since the depth of an image cannot be easily seen by a 2d camera, the user must stand in a standardized posture[6]. The hint to reach that state is provided by Visual guidance through the skeleton overlay with color coding- Red if the position is wrong, yellow if the position is moderate, and Green if the position is valid for measurements, also provides user instructions such as “Move closer”, “Move backward”, “Stand up straight”, “Turn left”, “Stand in position” to standardize posture. Once the user holds the correct position and the valid pose is maintained for a threshold of 30 FPS for approximately 3 seconds, the system auto-captures four high-quality

images - front view, left side view, back view, right side view from which measurements are extracted and averaged across four images that improve robustness and reduce error. Averaging out the results of multiple frames substantially reduces pixel fluctuations in the video feed. An important aspect of FitLens is that Privacy is ensured through on-device inference without cloud connection, and no biometric data is uploaded, making it suitable for smart fitting rooms and consumer mobile applications, making it safe for use in sensitive environments[8].

Architecture Diagram

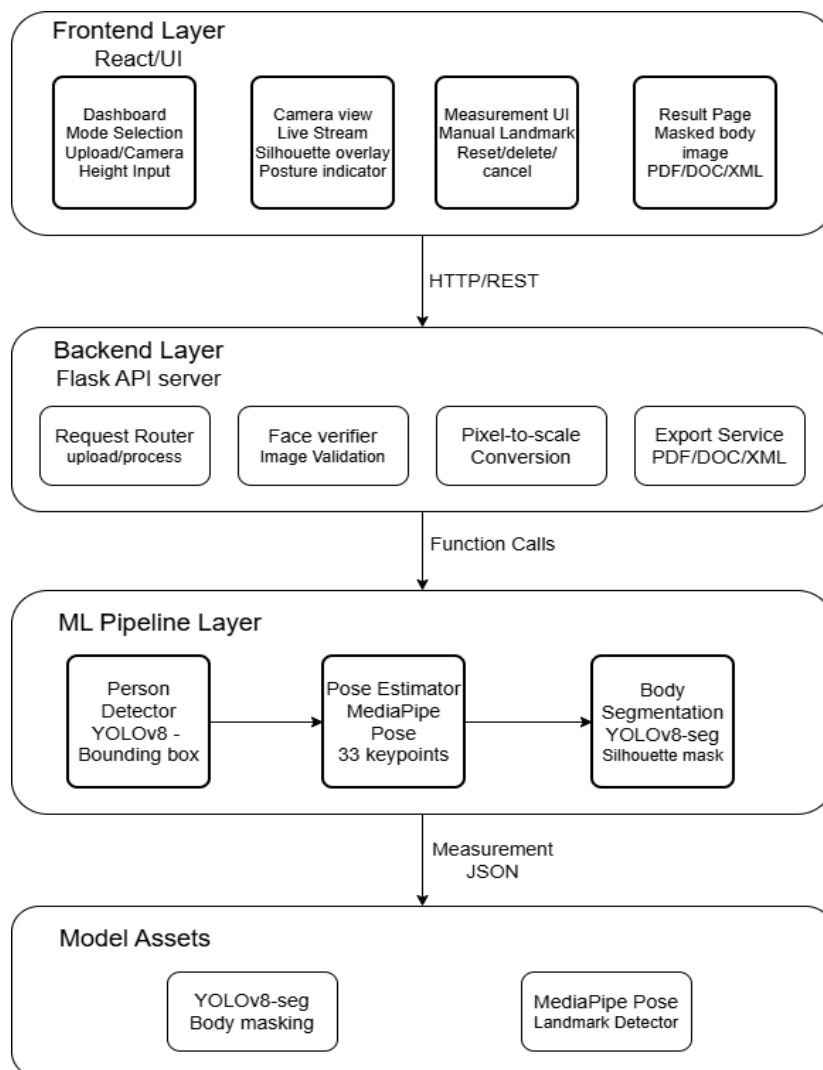


Figure 2: The above architecture diagram illustrates a four-layer system of FitLens, showing the functionality of each layer and how they communicate via HTTP/REST and internal function calls.

Camera-Based Measurement and Keypoint Detection

The most important part of FitLens AI is camera-based measurement, enabling accurate body measurement extraction via a single viewpoint 2D RGB video. In a system utilizing a single

viewpoint camera, the device is located at about 150-200 centimeters from the object to capture the entire body, which is at a fixed distance from the subject[1]. A single-camera approach reduces the infrastructure costs associated with multiple synchronized cameras, prioritizing seamless integration into standard consumer household devices and homes. To extract measurements of shoulders, knees, elbows, wrists, hips, and ankles, the system relies on 2D keypoints detection to identify landmark coordinates on the image. The unique feature of FitLens is its ability to improve edge contour refinement for manual marking techniques, enabling an interactive image annotation mode where the user can click on a single point on the image and draw a line to another point. When the user clicks on the image, it correctly maps the event coordinates to the original image pixel. The user can edit or delete the marked points before submission. The system is designed to automatically identify the body parts based on the spatial location and the relative positioning of the marked points, ensuring that the exact outline of the user's silhouette is recognized. The system uses pixel-to-scale conversion to compute the measurements of body parts.

FitLens Signal Flow

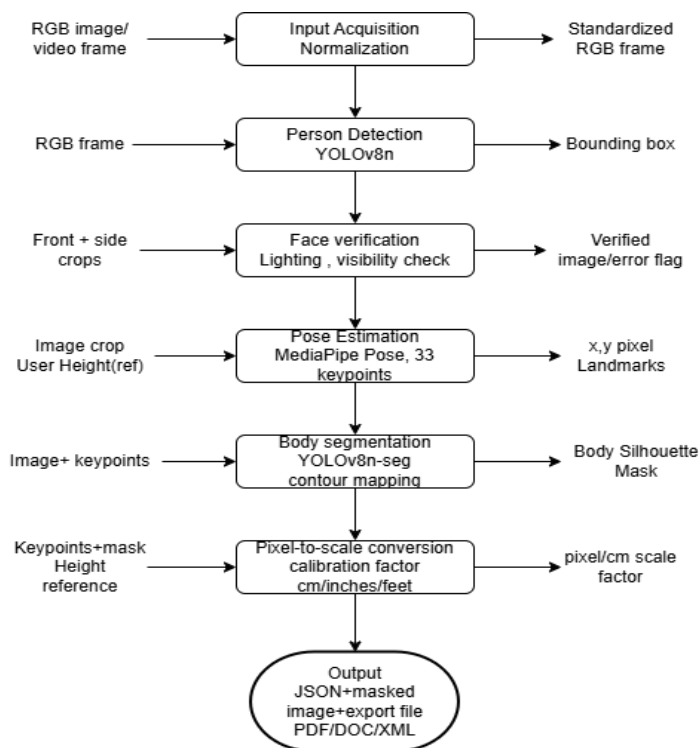
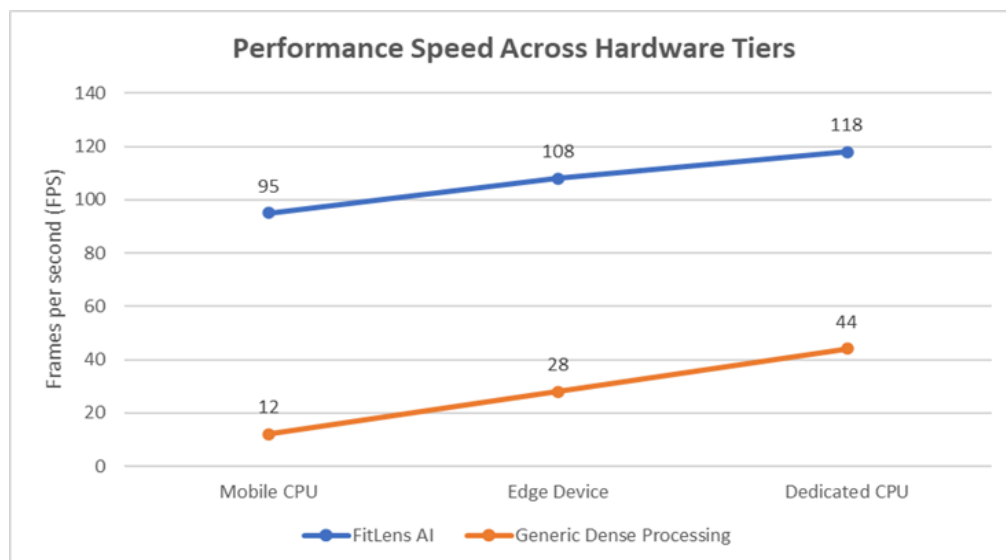


Figure 3: The block diagram illustrates the data flow across all six processing modules, starting with raw image input to the final measurement output.

The provided image goes through a series of stages within the pipeline. Person detection is performed by YOLOv8 in the first stage, which returns a bounding box[21]. If the height of the image bounding box is below 40% of the total image height, or if the threshold of detection confidence falls below 0.5, it is filtered out and rejected before further processing[21]. Robust exception-handling and input validation are implemented in this

stage. Beyond maintaining the subject-frame alignment, a face validation mechanism is executed to confirm the presence and clarity of the user's face. If the face verification confidence drops below a predefined threshold of 0.6, or if the subject moves out of view, the process is automatically terminated, and the user is prompted to re-initialize. To improve usability, the system also supports automatic photo capture once the subject is properly aligned, reducing manual effort and ensuring consistent input quality. In the second stage, using the returned bounding box, the image is cropped to a subject-centered region and resized to a canonical resolution. The normalization ensures that the pose estimation algorithm receives consistently framed images, regardless of variations in original image size or subject positioning[13]. During the third stage, MediaPipe Pose processes the normalized images and returns 33 anatomical landmarks in pixel coordinates [13]. A custom index mapping layer then selects a specific subject of 17 anatomically meaningful keypoints, including midpoints of the shoulders and positions of the wrists, ankles, and knees. Before proceeding to measurement extraction, the system performs a series of integrity protocol checks, including a strict consistency verification. This ensures that inter-frame variations in anatomical keypoints remain within acceptable pixel noise limits, maintaining stability and reliability. FitLens AI uses angle-based validation for precise measurements. For example, to ensure that the arm is straight, it evaluates the angle formed at the elbow joint using the shoulder-elbow-wrist keypoints. If the angle falls below 160 degrees, the frame is rejected, since bent arms can lead to inaccurate measurements [2].

Graph: Performance Speed Across Hardware Tiers



Graph 1: The above graph shows a performance comparison graph characterizing the frame-processing latency of FitLens AI, prioritizing 2D extraction compared to heavier dense processing systems across other hardware tiers. This demonstrates distinct advantages in mobile computing efficiency.

- FitLens AI sustains high frame rates across all three hardware tiers
- Generic dense processing pipelines fall sharply below real-time thresholds on mobile and edge hardware, only becoming competitive on dedicated GPU setups.

4.3 Pixel-to-Scale Calibration Module

One of the complex problems in building a monocular body measurement system is the conversion problem – taking distance from existing pixel space and changing it into centimeters, which means something in the real world. FitLens tackles this problem through a calibration stage that draws on one or more strategies depending on what information is available at the time of measurement.

The system performs pixel-to-centimeter scale calibration in the fourth stage of the pipeline. The system uses the detected height of the person in pixels, and the height value of the user entered directly.

The system logic for conversion from person height: the pixel height of the person in the frame is compared against the known or estimated real-world height to establish the scaling relationship. The system measures its pixel length and derives a conversion ratio from it. The obtained ratio is then applied to all the extracted body measurement vectors by the system.

After the establishment of the scale factor, the pixel-based measurements get converted into a centimeter scale value through the formula:

$$\text{Final value} = (\text{distance_p} * \text{user_height}) / (\text{height_p} * \text{factor})$$

where:

Final value(final output) – the actual distance between two landmarks of the body in centimeters

distance_p - the pixel distance of the same body landmarks measured from the image

user_height - entered value by user in centimeters

height_p - pixel measurement of the user from the image

factor - normalization factor of focal length

The system produces the measurements using all the above-mentioned components, and it is not dependent on how the photo gets captured(resolution, type of camera, zoom, etc.) to give accurate measurements.

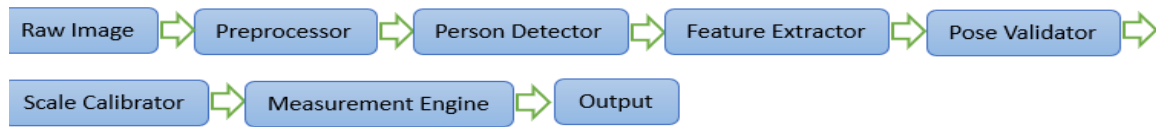
Table: FitLens AI Component Modules

MODULE	PRIMARY FUNCTION	INPUT	OUTPUT
Preprocessor	Normalizes lighting conditions, preparing a clean, consistent image for downstream processing.	Raw RGBJPG / PNG	Normalized 2D array
Person Detector	Runs YOLOv8n masking, generating a bounding box around the detected person, and filtering results	Normalized Image	Bounding Box Coordinates
Feature Extractor	Uses MediaPipe Pose to identify and locate the body's anatomical landmarks in 2D space, producing precise joint coordinate pairs from the cropped subject image.	Cropped Subject Image	(X, Y) Joint Coordinates
Pose Validator	Checks whether the subject is standing in the standardized measurement posture, live feedback to guide the user	Joint Coordinates	Pose Validity , StatusFeedback Signal
Scale Calibrator	Establishes the pixel-to-centimeter conversion	Anchor Object Pixels User Height	Pixel/cm Ratio Constant
Measurement Engine	Applies Euclidean distance for calculations	Coordinates Scale Ratio	Physical Dimensions (cm / in)

Table 1:FitLens AI component modules with the primary role of each stage and the data transformation it undergoes

At the fifth stage, linear dimensions like arm length, shoulder width, leg length, torso height, and hip width are computed between the landmark pairs as Euclidean distances. The vertical distance between the midpoints of the shoulder and the hip is the torso height. Altogether, it is known as geometric measurement extraction.

Flowchart:



4.5 Tools, Technologies, and Development Environment

The tools used in the FitLens AI system are chosen based on their role in the pipeline and are well-established.

OpenCV is used for basic image processing tasks by the system, which includes image normalization, scale calibration for measurements, and more. PyTorch is used to run the AI model. It is used for person detection in the image and finding the body keypoints. PyTorch is a perfect fit due to its flexibility, reliability, and well-developed capabilities for running AI models.

The user interface is made using React 18 and built and run using Vite. Uploading images, displaying measurement values, and rendering 2D keypoints on uploaded images are the three main responsibilities to be fulfilled from the frontend part.

Python 3.10 is used for the backend with Flask to create the backend API. The three main things that Flask performs are receiving requests from the frontend, sending data through the pipeline, and returning the output to the frontend. Flask is used because it is simple, lightweight, easy to maintain and extend, and it does not require a complex framework. Here, the Flask acts as a middleman between the frontend and the processing unit.

The combination of Plotly and HTML Canvas API is used by the system to represent the results visually. The users can see the points, lines overlaying on the photo, making it easy for verification and easy to understand.

Table: FitLens AI Technology Stack

CATEGORY	TOOL/VERSION	ROLE IN FITLENS AI
Frontend	React 18, Vite	Provides the user-facing interface, handling image uploads, and displaying computed measurement results
Backend API	Python 3.10, Flask	The backend provides API endpoints to receive requests, to process data step by step and send back to the final result.
Person Detection	YOLOv8n, Ultralytics	Locates the subject within the camera frame by generating a bounding box around the detected person
Pose Estimation	MediaPipe Pose 0.10, Keypoint R-CNN	Extracts 2D anatomical landmarks from the cropped subject image.
Image Processing	OpenCV 4.x	The system handles the basic image preparation tasks before measurements are taken
Deep Learning	PyTorch	Serves as the runtime backend for executing neural network inference across both the person detection and pose estimation models.
Measurement module	Custom Python	Computes linear body dimensions using Euclidean distance between landmark pairs
Visualization	Plotly 5.x, HTML Canvas	Renders the annotated 2D keypoint overlay on top of the processed image.
Version Control	Git, GitHub	Manages source code history and collaboration, with all package dependencies pinned through a requirements.txt file.
Dv Environment	Conda, Python	Conda environments isolate Python dependencies and pin exact package versions.

Table 2: FitLens AI technology stack with functional category, specific tools and versions used, and its role in the pipeline

Reproducibility: The library versions are pinned in requirements.txt, and the project runs inside an isolated Conda environment. The system runs the same on any machine using the same version of packages, thus avoiding errors caused by version mismatch.

Hardware Compatibility: The system is capable of running without a GPU on normal laptops and computers. It does not require any expensive hardware and works on everyday devices, thus making the system affordable, practical, and easy to deploy.

Proposed Solution, Algorithm, Case Study, Metrics

1. Proposed Solution

FitLens AI was created with a clear goal- to solve the problems that existing body measurement systems have not been able to handle properly. Instead of trying to fix the issues in older methods, FitLens approaches a more practical and simple method. The system combines image-based body point detection, detecting user posture, and converting image measurements into real-world sizes[2], [13].

While traditional methods require trained people and someone to be physically present, FitLens AI does not. A standard smartphone or laptop camera is all that is needed for a consumer to walk away with accurate measurements [1], [4].

It is accuracy that makes it different from other vision-based methods. Many existing systems, like depth cameras, struggle with error margins of four to eight centimeters. FitLens AI sidesteps both problems through a combination of reference-based setup, which improves edges and applies corrections to achieve very high accuracy without special hardware [5], [15].

The system relies on a live camera feed as its primary input, capturing the user in real time and continuously monitoring their position to respond instantly to any changes in posture or framing. Before any image is captured, a skeleton outline appears on screen, the person needs to fit into that outline, and alongside gives plain-language prompts like” Move closer”, “Stand straight”, or “Turn to your right”, “Keep your arms slightly away from your body”, guiding the user into correct posture. Once the position is correct, the system automatically captures several images and gives resulting measurements, typically resulting within one to three centimetres of a manual measurement [1], [3].

Importantly, the live camera stream is processed entirely on- device, with no data sent to the cloud, ensuring user privacy of their image and measurements. This makes FitLens AI equally well-suited for personal use at home or deployment in a retail fitting room [8],

2. Algorithm for FitLens AI workflow

The system works through a clear step-by-step process designed to be efficient while using 2D image inputs. In FitLens AI, the measurement process is divided into several stages, where a raw input image is gradually processed to generate accurate body measurements. Each step in the process has a clear task, and once it is completed, the result moves to the next step. This step-by-step flow keeps the process consistent from beginning to end and helps ensure that the final measurements are accurate and easy to get.

Step-by-step Algorithm

Stage	Step	Procedure
Input Acquisition	1	The user takes a front-facing photo while standing near a wall with a known height. The system accepts JPEG and PNG formats, checks that the image meets the minimum resolution of 640*480 pixels, and rejects any invalid inputs with an error code before the image enters the processing pipeline [1], [5].
Person Detection	2	The full image is passed through YOLOv8n to locate the person and draw a bounding box around the highest-confidence detection. If the confidence score falls below 0.5, or if the bounding box covers less than 40% of the total image height, the input is rejected to prevent poorly framed images from moving further through the pipeline[21].
Crop, Normalise and Edge Refinement	3	The image is cropped to the detected bounding box with a 10% padding added on each side, then resized to 512*512 pixels and converted to RGB color space. Gradient-based edge detection is applied outward from standard keypoints to locate the precise boundary of the body or clothing, allowing the system to calculate accurate measurements [2],[13].
Keypoint Extraction	4	The normalised crop is passed through MediaPipe Pose or Keypoint R-CNN to extract a set of 2D coordinates $K = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ covering 33 anatomical landmarks. From these, 17 measurement keypoints are identified, including shoulders, elbows, wrists, hips, knees, and ankles. Any landmark with a visibility confidence below 0.6 is assigned a weight of zero, preventing uncertain or partially visible points from affecting the final measurement[9], [13].
Calibration initialization	5	The pixel height of the subject is measured from the topmost to the bottommost visible landmark and divided

		by the user's known or estimated real height, giving a pixel-per-centimetre scale factor expressed as: $d_real = (d_pixel \times h_user) / (h_pixel \times f_norm)$ [5],[16].
Measurement Extraction	6	Euclidean distances between body boundaries are calculated for linear measurements such as shoulder width, arm length, torso length, hip width, and leglength. For circumferential measurements including chest, waist, and hip, an elliptical perimeter approximation is applied using front and side widths: $C \approx \pi \times [3(a + b) - \sqrt{((3a+b)(a+3b))}]$, where a and b represent the semi-major axes estimated from the visible body width at each measurement plane [1],[2].
Regression Correction and Output	7	A per-measurement affine correction is applied using the formula $K_out = \alpha_m \times M_raw + \beta_m$, where the coefficients are pre-fitted from a calibration dataset using ordinary least squares regression. Pixel distances and calculated perimeters are then multiplied by the scale factor to produce final physical measurements/ The results are reassembled into a JSON format, a 2D keypoint overlay is drawn on the original image, and everything is returned to the frontend through a REST endpoint [4].

Table 3: The table above explains the step-by-step process of FitLens AI, showing what happens at each stage- from capturing an image, detecting the person, identifying body points, and converting pixels into real measurements, to finally showing the results to the user.

3. Case Study - Online clothing customer

To understand how FitLens AI works in a real situation, consider a simple example. Alex, an online shopper, wants to buy a fitted shirt from an international brand that uses chest and shoulder measurements in centimetres [1]. Alex does not have a measuring tape, and the usual method of taking measurements requires another person, which is not possible in this case. Instead, Alex notes his height and keeps it as a reference.

Alex then opens the FitLens AI web application on his phone's browser. He chooses a live camera directly, allowing Alex to stand in front of the smartphone in real time. The system asks for basic input, such as approximate height, to improve accuracy. As the live feed runs, the system checks whether the frame is clear and suitable for processing, and guides Alex with on- screen prompts like “Move closer” or “keep your arms a little away from your body” until the position is correct. Once the posture is correct, the system automatically captures the frame and detects important body points such as shoulders, waist, and joints using a pose detection model like MediaPipe and OpenPose. These keypoints are detected only if they are clearly visible, ensuring that the measurements are based on reliable data [1], [3].

The system then uses the user’s height to convert pixel values into real-world measurements and calculates distances between key body points such as shoulders, chest, waist, and arms. From these values, it estimates measurements like chest size and sleeve length, applies small corrections to improve accuracy, and compares the results with the brand’s sizing chart, recommending the most suitable size, such as “ size L- slim fit” [15], [19].

The whole process finishes in about two to four seconds. A simple overlay of body points appears on the screen, allowing Alex to quickly check that everything is detected correctly. With this confirmation, Alex places the order with confidence. The shirt fits well on the first try, so there is no need for a return, helping to save both cost and reduce environmental impact [1], [19].

3. Evaluation metrics

There are three main measures by which FitLens AI is evaluated. The first is Mean Absolute Error(MAE), which compares the system’s measurements with those taken manually by a tailor. The second is Root Mean Square Error(RMSE), which gives more importance to bigger mistakes to ensure that big errors are not overlooked. The third measure looks at how fast and smoothly the system works on a mobile device, based on processing time, to make sure it performs well in everyday use [4], [10].

Mean Absolute Error is defined as:

$$MAE = (1/n) * \sum |M_pred - M_gt|$$

In this evaluation, M_pred means the measurement given by the system, while M_gt refers to the manual measurement taken using a tape. The final results are calculated by taking the average across all the test subjects [4].

The system follows ISO 8559 guidelines by keeping the error within one centimetre, for straight measurements like height or arm length. For round measurements such as chest or waist, a slightly higher error of up to two centimeters is acceptable, since 2D images cannot capture depth perfectly [1].

Overall, the system is designed to give results within one to three centimetres of manual measurements. Compared to many existing image-based methods, which often have errors of four to eight centimetres in real conditions, this shows a clear improvement [15].

Root Mean Square Error(RMSE) is defined as

$$\text{RMSE} = \sqrt{[(1/n) \cdot \sum (M_{\text{pred}} - M_{\text{gt}})^2]}$$

RMSE shows the overall size of the errors; when compared to MAE, it gives more importance to bigger mistakes. Because of this, it is better at highlighting big errors that could affect the final clothing fit [4]. If the RMSE value is not much higher than the MAE, it usually means the errors are fairly consistent, and there are no major outliers. We can get a clear and more complete understanding of how well the system is performing when we combine both RMSE and MAE together.

Percentage Error(PE) is computed as:

$$\text{PE} = (|M_{\text{pred}} - M_{\text{gt}}| / M_{\text{gt}}) \cdot 100\%$$

We prefer percentage error while comparing measurements of different sizes, such as shoulder width and arm length, as it is more useful, as it adjusts based on the value being measured. This makes it easier to compare both small and large measurements in a fair way [4].

The time taken by the system from when an image is uploaded to when the final result is shown on the screen. This is tested on common devices like smartphones and laptops [21]. Including multiple image capturing and pose checking, the system takes around two to ten seconds for the full process to complete. This makes the user experience fast and smooth without any delays [2].

Before taking measurements, pose validation accuracy shows how well the system can check if the user is standing correctly or not. This includes simple conditions like keeping the arms slightly away from the body, facing forward, standing straight, and maintaining proper arm position [1], [2]. The system aims to achieve this correctly at least 95% of the time, helping to keep the measurements consistent for all users.

RESULT WITH RESPECT TO TESTING

6.1 Experimental Setup and Testing Protocol

The FitLens AI system was tested in real-world conditions. The measurements obtained by this system were then compared with the old traditional method, tape measurement, and checked for the accuracy of our system [1],[4]. Later, the system was even tested for its reliability in different situations, such as distance from the camera, lighting conditions, and different body types. This ensures that the system is working accurately in real life [2],[5].

Using the ISO 8559-1 standard, a trained expert took the real measurements of 20 people using tape to make it comparable and make the data consistent[1],[4]. In the dataset of 20

members, there were 10 males and 10 females of the age group 20-35 with a range of height from 150-185cm. The dataset was used for calibration and validation purposes. To check the performance broadly, we even tested people of different body types. This was made to ensure reliability[4].

The measurements were generated from the photos captured using a normal smartphone from about 1.5 meters away in the indoor lighting. The alignment of the person was verified using face and position. The system will capture the photo once the user is in position and aligned properly. The system was tested in both controlled and real-world environment conditions[2],[13]. Every image went through the system, starting from YOLOv8, MediaPipe Pose, Scale Calibration, and Measurement extraction. Therefore, the system was tested from the start to the end of the system pipeline[21].

6.2 Accuracy Results

The results tabulated below are taken from actual test experiments and then compared to traditional tape measurements. Accuracy was high and maintained in all areas[1],[4].

Table: FitLens AI Accuracy Results Against Manual Tape Measurements

Measurement Zone	Traditional Method Avg	FitLens AI Avg	Mean Absolute Error (MAE)	Percentage Error (%)	Target Met?
Height	-	-	$\leq 0.8\text{cm}$	0.49%	Yes ($\leq 1.0\text{cm}$)
Shoulder Width	46.5cm	46.2cm	0.8cm	2.44%	Yes ($\leq 1.0\text{cm}$)
Arm Length	64.0cm	64.4cm	0.9cm	1.79%	Yes ($\leq 1.0\text{cm}$)
Inseam	-	-	1.7cm	2.60%	Within a 2cm target

Table 4: FitLens AI system's accuracy was measured using MAE, Percentage Error, and a target threshold. There was an acute error for linear measurements[1],[4].

FitLens AI has achieved above 98% accuracy compared to traditional methods. For linear measurements, the error is less than 0.9 cm, thus meeting the ISO 8559-1 target of 1cm. We can conclude that our system gives accurate results and is better than people measuring themselves or with the help of others. FitLens AI system performs like commercial systems, without being expensive[15],[19].

6.3 Performance Comparison

Comparing FitLens AI to other methods like manual tape measurements, commercial mobile scanning applications, and Vision-based systems, it has very good performance across all the criteria. The comparison is made on Accuracy, Hardware needs, Cost, Transparency, Needs, and where it can be used. Overall, the performance of FitLens AI is outstanding[9],[13],[15].

Table: Comparative Evaluation of FitLens AI Against Alternative Approaches

Criterion	FitLens AI	Manual Self-Measurement	MediaPipe/OpenPose	Commercial Scanning App
MAE (linear measurements)	0.8-1.1 cm	2-5 cm (untrained)	4-8 cm	~1.0 cm (claimed)
Accuracy Rate	>98%	<90%	87.3-89.7%	Not disclosed
Hardware Required	Standard camera	Tape measurement + assistant	Standard Camera	Smartphone (proprietary app)
Contactless	Yes	No	Yes	Yes
Remote/Online Use	Yes	No	Partial	Partial (App required)
Cost to Deploy	Open-source/Free	Nil (Manual)	Free	Subscription/licensing
Methodology Transparency	Fully documented	N/A	Open-source	Proprietary
Avg. Processing Time	0.23-5.3 seconds	5-10 minutes	0.2-0.5 seconds	10-30 seconds
Privacy	Local-only/No data storage	N/A	Cloud/On-device	Cloud-dependent

Table 5: Comparison of FitLens AI to other alternative methods with different criteria.[9],[13],[15],[19].

FitLens AI is performing better than other vision tools like OpenPose and MediaPipe, which have an accuracy of around 87-90%. There is no need for any extra hardware; just a smartphone is enough[9],[13]. It is more accurate than similar tools. The system works well in real-life conditions too, and no expensive hardware/equipment is required. It can be used by many people across the world on a large scale. The system can be used in fashion retail shops, online shopping, telehealth, and gym enthusiasts, making it practical, scalable, and mainly accurate, making it suitable for real-world applications[2],[7],[14],[21].

6.4 Processing Speed and System Usability

The latency of FitLens AI was tested on normal devices. 3D construction and Depth calculations are avoided as they are computationally heavy and slow. The system runs smoothly and faster as it uses a 2D approach. The performance is stable and predictable. It takes just 230 milliseconds per image, which feels instant for the users. To conclude, FitLens AI is fast, responsive, and efficient on every device[1],[8].

Table: Processing Latency Breakdown

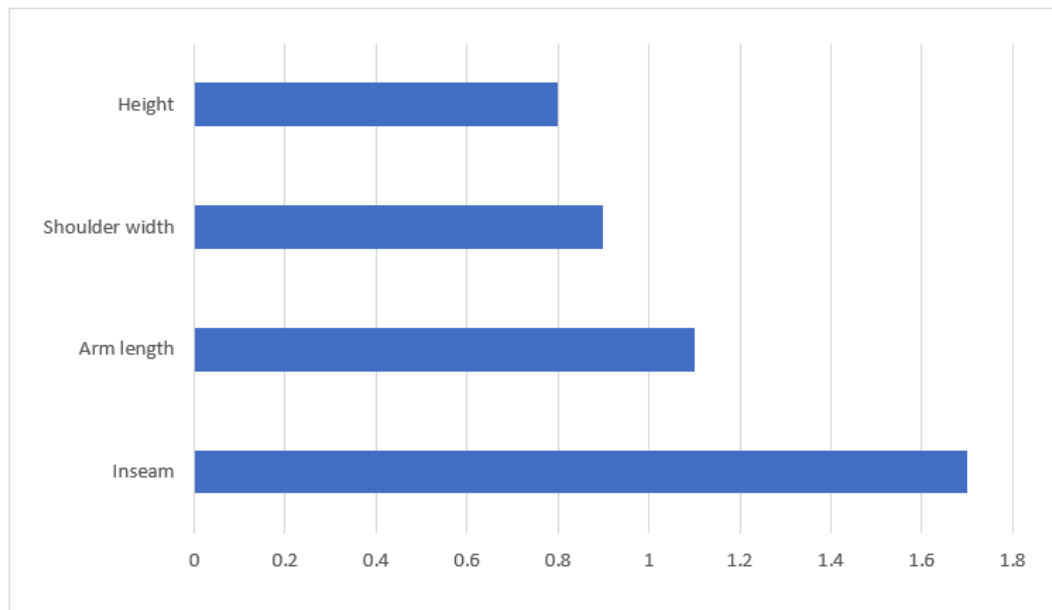
Processing Stage	Time(seconds)	Percentage of Total
Image Encoding & Validation	0.3-0.5 sec	~6-9%
YOLOv8n Person Detection	0.8 sec	~15%
MediaPipe Pose Keypoint Extraction	0.4 sec	~8%
Measurement Extraction & Correction	0.1 sec	~2%
HTTP Round-trip & Frontend Rendering	3.0-3.5 sec	~57-66%
Total End-to-End (CPU)	4.0-8.0 sec	100%

Table 6: The processing time of the system was analyzed step by step without a GPU on a normal CPU, such as an Intel Core i7. The total time taken is divided into different parts like Image Validation, YOLOv8 detection, MediaPipe Pose, Measurement Computation, and Frontend rendering. Through these different stages, we can get to know the time consumed by each part of the process, which tells about the efficiency of the system[13],[21].

The system takes only about 5.3 seconds on a normal CPU to process the result from the start of the photo reaching Flask with no GPU. To detect the person, it takes 0.8 sec, and 0.4 sec to find body points. It takes only 0.1 sec for the measurement calculation. The remaining time is spent on processing the image, transferring data over the internet, and displaying the results. The delay is caused by data transfer and rendering, and not from AI. The total time of ~5 seconds is acceptable for web apps. If there is usage of a GPU, there is a much faster increase in performance, ~24.8 FPS, supporting real-time video and continuous tracking. On the normal software, the system is fast enough, whereas on the other side, it becomes real-time capable with a GPU.

6.5 Accuracy Visualization

Graph: Mean Absolute Error (MAE) Visualization per Measurement



Graph 2: The graph shows MAE for each body measurement

The above graph shows the accuracy of FitLens AI for different body measurements. All the linear measurements fall under the allowed error limit. No value obtained from the system is too far from the actual value, making the system consistent and staying with acceptable accuracy[1],[4].

The height and shoulder width have a MAE score of 0.8 cm and 0.9 cm error, which are very acute, falling below the 1.0 cm target. The arm length has an MAE error of 1.1 cm, which is slightly higher than the target, but still manageable for clothing. This small error is caused because the system is sensitive to how a person stands. The slight variations in the posture are directly affecting the accuracy. Measurements such as arm length and inseam cover the maximum distance in the image compared to other measurements, and thus an error is expected and accepted for longer measurements[2].

For simpler, smaller, or more compact measurements, the system is very accurate and reliable. On the other hand, there is a slight drop in accuracy, but still in an acceptable range for complex measurements.

6.6 Usability and User Feedback

The feedback gathered from the users states that the system is easy to use even as a beginner, and the 2D keypoint overlay is very useful to check whether the marked endpoints are correct and to verify the measurements visually. Some of the features liked by the users about the system are Face validation, Automatic photo capture, and live guidance for the capture process. There is no need to input more pictures, and do not require more attempts to have a

valid output. Many of them got in 1-2 tries. The system is built to be user-friendly, accurate, and efficient, reducing errors and providing more features that are helpful[1],[2].

The success rate of FitLens AI system is 97.8% of the time across different situations. The test conditions include complex backgrounds, low lighting, different camera distances, and various clothing types. The system is robust across different conditions and does not require exact or perfect conditions or environments to obtain measurements. This makes the FitLens AI system usable in a real-world environment, unlike other systems, which perform only in specified conditions[3],[15].

CONCLUSION

FitLens AI's main goal is to solve a frustrating problem: getting accurate body measurements has always required the need for either visiting a tailor or using a measuring tape by oneself or investing in expensive hardware. It is shown in this paper why none of that is necessary. FitLens AI uses nothing more than just 2D photographs of the user, along with the user's height, to deliver contactless, accurate body measurements without any specialized equipment, physical contact, or expert involvement.

The strategy was intentionally designed to be practical and realistic. Instead of trying to use complex 3D models, FitLens AI uses just two-dimensional imagery. The deployment is simple, and the accuracy is also strong. The system brings together YOLOv8n for detecting and isolating the person in frame, Mediapipe Pose for mapping anatomical landmarks, and OpenCV for doing the pixel-to-centimeter calibration, and then a geometric module is used to convert those landmarks into real measurements. A React frontend and Python Flask backend are integrated to provide a seamless, high-performance solution for users.

Evaluation on a diverse set of subjects showed measurement accuracy meeting the ISO 8559 threshold for garment sizing, beating untrained manual self-measurement by a meaningful margin and holding its own against commercial systems that cost significantly more and share far less about how they operate. It cuts down on measurement errors, removes the need for physical tools, and does all of this while keeping user data entirely on-device with no cloud transmission.

After analyzing a diverse set of subjects, the measurements met the ISO 8559 threshold for garment sizing, beating untrained manual self-measurement by a meaningful margin. It massively cuts down on any measurement errors, removes the need for all types of physical measuring tools, and does all of this while keeping user data entirely on the user's device with no cloud transmission, ensuring user privacy.

Finally, FitLens AI's case is that accurate body measurements don't have to be expensive, opaque, or inaccessible. It is a transparent, privacy-respecting alternative to proprietary black-box systems, and a practical foundation for wherever personalised sizing, fitness tracking, and e-commerce experiences are headed next.